

Complex Velocity Potential in Fluid Flow

Overview

- ▶ Goal: describe 2D fluid flow using complex analysis
- ▶ Focus: incompressible and irrotational flow
- ▶ Key ideas:
 - ▶ Velocity field $\mathbf{v}(x, y)$
 - ▶ Divergence $\nabla \cdot \mathbf{v}$
 - ▶ Curl $\nabla \times \mathbf{v}$
 - ▶ Velocity potential ϕ and stream function ψ
 - ▶ Complex potential $f(z) = \phi + i\psi$

Velocity Field Description

A fluid flow is described by a velocity field

$$\mathbf{v}(x, y) = (u(x, y), v(x, y))$$

- ▶ u = horizontal velocity component
- ▶ v = vertical velocity component
- ▶ Field assigns a velocity vector to each point

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Interpretation:

- ▶ Tracks motion of fluid particles
- ▶ Fundamental object in fluid mechanics

Flux and Divergence (Physical Derivation)

Consider a small rectangular box with sides Δx , Δy .

Flux in x -direction:

$$u(x + \Delta x, y) - u(x, y)$$

Flux in y -direction:

$$v(x, y + \Delta y) - v(x, y)$$

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$$\approx \frac{\partial u}{\partial x} \Delta x \Delta y + \frac{\partial v}{\partial y} \Delta x \Delta y$$

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Divide by area:

$$\nabla \cdot \mathbf{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

Physical Meaning of Divergence

- ▶ Measures **net outflow per unit area**
- ▶ Interpretation:
 - ▶ > 0 : source (fluid created locally)
 - ▶ < 0 : sink (fluid removed)
 - ▶ $= 0$: no sources or sinks

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$$\nabla \cdot \mathbf{v} = 0$$

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Physical meaning:

- ▶ Fluid density is constant
- ▶ Volume is preserved

Circulation and Curl (Physical Derivation)

Consider a small rectangular loop.

Circulation around boundary:

$$\oint \mathbf{v} \cdot d\mathbf{r}$$

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Divide by area:

$$(\nabla \times \mathbf{v}) \cdot \mathbf{k} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Physical Meaning of Curl

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Irrotational flow:

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Interpretation:

- ▶ No local spinning motion
- ▶ No vortices

Velocity Potential

If flow is irrotational:

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Then there exists a scalar function ϕ such that:

$$\mathbf{v} = \nabla \phi$$

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That is:

$$u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}$$

Laplace Equation

Combine with incompressibility:

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Conclusion:

- ▶ Velocity potential satisfies Laplace's equation
- ▶ Flow is governed by harmonic functions

Stream Function

Define $\psi(x, y)$ such that:

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Then:

- ▶ $\psi = \text{constant}$ are streamlines
- ▶ Automatically satisfies incompressibility

Cauchy-Riemann Equations

Comparing:

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Conclusion:

- ▶ ϕ and ψ satisfy Cauchy-Riemann equations

Complex Velocity Potential

Define:

$$f(z) = \phi(x, y) + i\psi(x, y)$$

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$$f(z) = \phi(x, y) + i\psi(x, y)$$

Then:

- ▶ f is analytic
- ▶ Its derivative gives velocity:

$$f'(z) = u - iv$$

Physical Interpretation

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Key properties:

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- ▶ $\psi = \text{const} \rightarrow$ streamlines
- ▶ Orthogonal families

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Complex analysis encodes fluid flow geometry

Summary

- ▶ Divergence $\rightarrow$ flux density (compressibility)
- ▶ Curl $\rightarrow$ local rotation
- ▶ Incompressible: $\nabla \cdot \mathbf{v} = 0$
- ▶ Irrotational: $\nabla \times \mathbf{v} = 0$
- ▶ $\mathbf{v} = \nabla \phi$
- ▶ ϕ harmonic
- ▶ Complex potential $f = \phi + i\psi$